From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

FIITJEE

JEE(Advanced)-2014

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
CRT–I
(Paper-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**C**	**C**	**D**
2.	**A**	**D**	**A**
3.	**B**	**D**	**B**
4.	**B**	**C**	**A**
5.	**A**	**B**	**B**
6.	**D**	**D**	**A**
7.	**B**	**C**	**C**
8.	**A.**	**C**	**A**
9.	**B**	**B**	**C**
10.	**C**	**B**	**A**
11.	**A, B, C**	**A, C**	**A, B, C**
12.	**A, B, C, D**	**B, C**	**A, B, C, D**
13.	**B, C, D**	**A, C**	**A, D**
14.	**A, B, C**	**A, B, C**	**B, C**
15.	**A, B, C**	**A, C**	**B, C**
1.	**2**	**2**	**4**
2.	**1**	**1**	**2**
3.	**5**	**4**	**6**
4.	**1**	**3**	**2**
5.	**2**	**6**	**3**

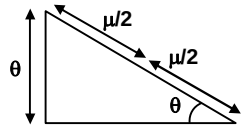
Physics

PART – I

SECTION – A

1. $U_{rel} = g \times t = 10 \times 1$
   = 10 m/s
   $S_1 = \frac{1}{2} g(1) = 5m$
   $S_{rel} = S_1 + u_{re\ell} + \frac{1}{2} a_{re\ell} t^2 = 5 + 10 \times 3 + 0 = 35$ m

2. $\sin\theta = \frac{h}{\ell}$
   $mgh = \mu\, mg \cos\theta(\ell/2)$
   $\frac{h}{\ell} = \frac{\mu \cos\theta}{2}$
   $\tan\theta = \frac{\mu}{2}$



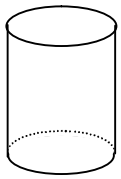
3. Centre of mass does not move in the absence of external force so
   $m_1x_1 = m_2x_2$
   $\frac{x_1}{x_2} = \frac{m_2}{m_1}$

4. $A = \frac{dA_1}{dt} = \frac{r^2}{2}\frac{d\theta}{dt} = \frac{r^2}{2}\omega$
   $L = Mr^2\omega$
   L = 2MA

5. $F = \frac{Gm(M-m)}{r^2} = 0$
   $\frac{dF}{dm} = \frac{G}{r^2}(M - 2m) = 0$
   M = 2m
   m = (M/2)

6. $\frac{\rho A \ell g}{A} = s$
   $\ell = \frac{s}{\rho g}$

7. $\rho gh\pi r^2 = \int_0^h (2\pi r dh)\rho gh$
   $\rho gh\pi r^2 = 2\pi r\rho g\frac{h^2}{2}$
   $r = 2\frac{h}{2}$
   r = h



8. Avg. speed $3 = \dfrac{\sqrt{(v_r t)^2 + (v_{mr} t)^2}}{t}$

$\Rightarrow v_r^2 + 5 = 9$

$\Rightarrow v_r = 2 m/s$

9. $v_{avg} = \dfrac{\frac{1}{2} \times \frac{t}{2} \times v + \frac{1}{2} \times v}{t} = \dfrac{3v}{4}$

10. $\dfrac{m}{2} g - T = \dfrac{m}{2} a$ …(i)

$T \cos 60° = \dfrac{ma}{\cos 60°}$ …(ii)

Solving (i) and (ii) acceleration of ring = $\dfrac{2g}{9}$

11. $mg - \dfrac{2kq^2}{R^2} + T = \dfrac{mv^2}{R}$ …(i)

$-mgR + \dfrac{1}{2} mu^2 = mgR + \dfrac{1}{2} mv^2$ …(ii)

if $T = 0 \Rightarrow u_{min} = \sqrt{5gR - \dfrac{2Kq^2}{mR}}$

12. $F = -\dfrac{dU}{dx} = -5(2x - 4)$

At mean position $F = 0 \Rightarrow x = 2m$

$U_{min} = -20$ J

$a = -50 \times 2(x - 2)$

$\omega = 10$ rad/sec

$T = \pi/5$ sec

13. If the volume immersed initially is (V/3). Then $\dfrac{V}{3} \rho g = mg$ …(i)

If the volume immersed when the system accelerates is V′ then

$V' \rho \left(g + \dfrac{g}{2} \right) - mg = \dfrac{mg}{2} \Rightarrow V' = \dfrac{V}{3}$

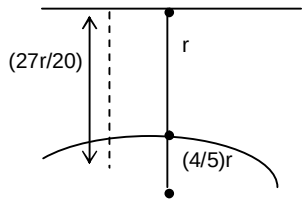
14. The potential of the two surface will be equal when the whole charge Q flows from inner to other shell.

SECTION –C

1. Consideration refraction at glass-water interface

$\dfrac{\mu_2}{v} - \dfrac{\mu_1}{u} = \dfrac{\mu_2 - \mu_1}{R}$

$\Rightarrow \dfrac{4}{3v} - \dfrac{3}{-2r} = \dfrac{(4/3) - (3/2)}{r}$



$$\Rightarrow \frac{4}{3v}+\frac{3}{2r}=-\frac{1}{6r}$$

$$\therefore\ v=-\frac{r}{5}$$

Now refraction at water air surface

$$u=-\left(r+\frac{4}{5}r\right)=-\frac{9r}{5}$$

$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{\infty}$$

$$\Rightarrow \frac{1}{v}+\frac{4\times5}{3\times9r}=\frac{1-(4/3)}{\infty}$$

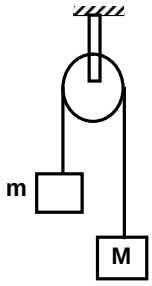
$$\frac{1}{v}=-\frac{20}{27r}$$

$$v=-\frac{27r}{20}$$

so height above centre $= 2r - \frac{27r}{20}$

$$=\frac{40r-27r}{20}=\frac{13}{20}r = 13 \text{ cm.}$$

2. $Mg - T = Ma$ …(i)
$T - mg = ma$ …(ii)



$$T=\frac{2Mmg}{(M+m)}$$

$$\Rightarrow \frac{T}{A}=\frac{2Mmg}{A(M+m)}=\frac{2mg}{A\left(1+\frac{m}{M}\right)}=2\times10^9$$

$\Rightarrow M = 1.86$ Kg

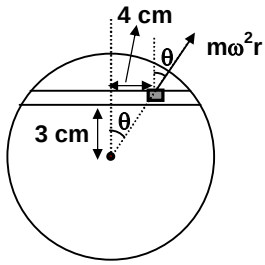
3. Relative motion between block and table will start when

$m\omega^2 r \sin\theta = \mu(mg + m\omega^2 r\cos\theta)$ …(i)

$\omega = \alpha t$ …(ii)

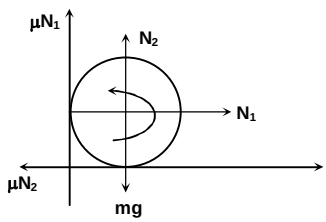
solving (i) and (ii)

$t = \sqrt{500} = 22.4$ sec .



4. $\omega = \sqrt{\frac{\mu g}{R}} = 1$ rad/sec

5. $\mu N_1 + N_2 = mg$ . . . (1)
$N_1 = \mu N_2$ . . . (2)



$$N_2=\frac{mg}{\left(1+\mu^2\right)}$$

$$N_1=\frac{\mu mg}{\left(1+\mu^2\right)}$$

Torque about centre of mass

$\mu(N_1 + N_2)R = \frac{1}{2}mR^2\alpha$

$\alpha = \frac{2\mu(N_1+N_2)}{mR} = \frac{2\mu(1+\mu)g}{(1+\mu^2)R}$

$\omega^2 = \omega_0^2 - 2\alpha\theta$

$\theta = \frac{\omega_0^2}{2\alpha}$

$\theta = \frac{\omega_0^2(1+\mu^2)R}{2\times 2\mu(1+\mu)g}$

Hence number of turns

$N = \frac{\theta}{2\pi} = \frac{\omega_0^2R(1+\mu^2)}{4\mu(1+\mu)g.2\pi}$

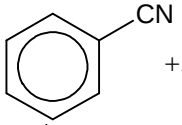
$N = \frac{\omega_0^2R(1+\mu^2)}{8\mu\pi g(1+\mu)}$

Chemistry

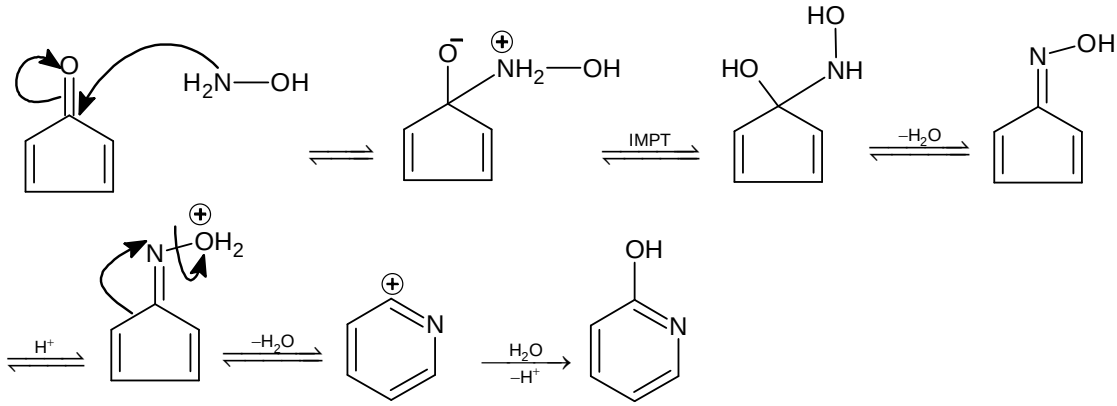
PART – II

SECTION – A

1. In 'A' π bond is at bridge head which is not stabilized in the given structure;

In 'B' the product is  $+ACOH$;

In 'D' the product is 

2. 

5. CH_3Cl is formed by S_N2 while $CH_3CHClC_2H_5$ is formed by S_Ni.

6.

	$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$		
Let initially no. of moles	x	–	–
at equilibrium(volume V)	$x(1-\alpha)$	$x\alpha$	$x\alpha$

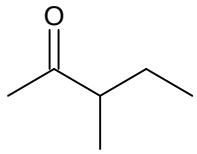
$K_c = \dfrac{x^2\alpha^2}{x(1-\alpha)V}$, also $K_p = K_c(RT)^{\Delta n_g}$

on reducing volume to $\dfrac{V}{2}$, initial concentration of PCl_5 doubles and its degree of dissociation is increased but K_p does not change, as it is characteristic constant for a reaction at constant temperature.

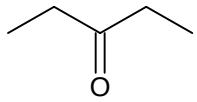
7. $Al_2Cl_6 + 12H_2O \longrightarrow 2H_3\left[Al(OH)_6\right] + 6HCl$;

10. In blast furnace,
In combustion zone, CO is produced ultimately which reduce Fe_2O_3 to Fe_3O_4 in reduction zone, which is uppermost zone.

11. P =

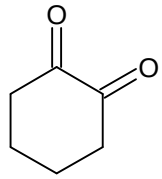


Q =



R = $CH_3COOH + C_2H_5OH$; $C_2H_5OH \xrightarrow{\text{Oxidation by } XO^-} CH_3CHO$

S =



12. Due to hydrogen bonding Gauche conformation of $H_2C(NH_2)—CH_2—NH_2$ and $H_2C(OH)—CH_2F$

are more stable than their anti-conformations.

In case of $ClCH_2CH_2Cl$, on increasing temperature, % of Gauche conformation increases. Hence dipole moment increases. In case of option (D) boat conformation is most stable.

13. Mn appears colourless in reducing flame in Borax Bead Test.

$Fe^{3+} + K_4\left[Fe(CN)_6\right] \longrightarrow \underset{\text{Prussian blue}}{Fe_4\left[Fe(CN)_6\right]_3} \downarrow + K^+$

$Na_2S_2O_3.5H_2O \xrightarrow{\Delta} Na_2S_5 + Na_2SO_4 + H_2O$

14. At pH = pI, amino acid exists as Zwitter Ion, they are highly soluble in polar solvents.

15. pH of 10^{-8} M HCl (aq) is 6.96 and pH of 10^{-8} M NaOH (aq) is 7.04

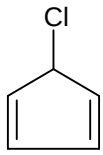
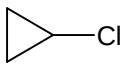
SECTION – C

1. $\pi = CRT \times i$
hence i = 1.1; so $\alpha = 0.1$
hence $[H^+] = c\alpha = 0.1 \times 0.1 = 10^{-2}$ M
pH = 2

2. It has only one CH_2OH group.

3. Three equivalents to remove three hydrogens from CH_3 group and one for attack on C of CHO group.

4. $BaCrO_4$ – Yellow, soluble in dil HNO_3
Hg_2CrO_4 – Red, soluble in conc. HNO_3
ZnS – White, soluble in Conc. HNO_3
$BaSO_4$ – White, insoluble in all mineral acids
BaS_2O_3, CH_3COOAg and $AgNO_2$ all are white solid and are soluble in dilute HNO_3 solution.

5. Except  and 

Mathematics

PART – III

SECTION – A

1. $[y + [y]] = 2\cos x \Rightarrow [y] = \cos x$
$y = \frac{1}{3}[\sin x + [\sin x + [\sin x]]] = [\sin x]$
$\Rightarrow [\sin x] = \cos x$
Number of solution in $[0, 2\pi]$ is 0
Hence total solution is 0.
$\therefore$ both are periodic with period 2π

2. $(x - 0)^2 + (y - k)^2 = k^2$
$\Rightarrow x^2 + (y - k)^2 = k^2$
$2x + 2(y - k)\frac{dy}{dx} = 0$
$\Rightarrow \frac{dy}{dx} = -\frac{x}{y-k}$
$y - k = -\frac{xdx}{dy}$
$k = y + \frac{xdx}{dy}$

$$\Rightarrow x^2 + \left(y - \left(y + \frac{xdx}{dy}\right)\right)^2 = \left(y + \frac{xdx}{dy}\right)^2$$

$$\Rightarrow x^2 + x^2\left(\frac{dx}{dy}\right)^2 = y^2 + x^2\left(\frac{dx}{dy}\right)^2 + \frac{2xydx}{dy}$$

$$x^2 = y^2 + \frac{2xydx}{dy}$$

$$(x^2 + y^2)\frac{dy}{dx} - 2xy = 0$$

3. Given $\alpha < \beta < \gamma < \delta$ also $\sin\alpha = \sin\beta = \sin\gamma = \sin\delta = k$
and $\alpha, \beta, \gamma, \delta$ are smallest positive angles
$\therefore \beta = \pi - \alpha, \gamma = 2\pi + \alpha, \delta = 3\pi - \alpha$
as $\sin\beta = \sin\alpha$ and $\beta > \alpha$
$\sin\beta = \sin\gamma$ and $\gamma > \beta$
$\sin\gamma = \sin\delta$ and $\delta > \gamma$
Putting these values in the given expansion, we have given expression

$$= 2\left(\sin\frac{\alpha}{2} + \cos\frac{\alpha}{2}\right) = 2\sqrt{1+\sin\alpha} = 2\sqrt{1+k}$$

4. $x^3 - x < -a^2 + a - \frac{2}{3\sqrt{3}}$

$f(x) = x^3 - x$
$f'(x) = 3x^2 - 1 = 0$

$$x = \pm\frac{1}{\sqrt{3}}$$

In positive region minimum value of f(x) = $\frac{1}{3\sqrt{3}} - \frac{1}{\sqrt{3}} = -\frac{2}{3\sqrt{3}}$

So, $-a^2 + a - \frac{2}{3\sqrt{3}} > -\frac{2}{3\sqrt{3}}$

$a^2 - a < 0$
$a \in (0, 1)$

5. $\underset{v}{\sqrt{3x-2}} + \underset{u}{\sqrt{4x+4}} = \underset{p}{\sqrt{5x+1}} + \underset{q}{\sqrt{4x-5}}$ …(1)

$u^2 - v^2 = x + 6 = p^2 - q^2$

$u - v = p - q$ …(2)

solving (1) and (2)
we get, $2\sqrt{4x+4} = 2\sqrt{5x+1}$
$\Rightarrow x = 3$

6. Limit be equal to y

$$\log y = \frac{1}{n}\lim_{n\to\infty}\left[\log\left(\frac{n+1}{n}\right)^{\lambda} + \log\left(\frac{n+2}{n}\right)^{\lambda} + \ldots\right]$$

$$= \frac{1}{n}\lim_{n\to\infty}\sum_{r=1}^{n}\left[\log\left(\frac{n+r}{n}\right)^{\lambda}\right] = \frac{1}{n}\lim_{n\to\infty}\sum_{r=1}^{n}\lambda\log\left(1+\frac{r}{n}\right) = \lambda\int_0^1\log(1+x)\,dx$$

$= \lambda \cdot 2\left[\log 2 - \frac{1}{2}\right] = \lambda[\log 4 - \log e] = \lambda \log\left(\frac{4}{e}\right)$

$\log y = \log\left(\frac{4}{e}\right)^{\lambda}$

7. $0 \le \arg z \le \frac{\pi}{4}$, represent the region of complex plane lying in the first quadrant and bounded by x-axis and the line y = x
$|2z - 4i| = 2|z - 2i|$
least value of $|z - 2i|$ is length of perpendicular from (0, 2) to y = x, which is $\sqrt{2}$
So the least value of $\sqrt{2}\,|2z - 4i|$ is 4

8. Let $f(x) = 2x^3 + 3x^2 - 12x + 3$
$\Rightarrow f(x) = 0$ has three real roots (α, β, γ)
$\alpha + \beta + \gamma = -\frac{3}{2}$
Centroid $= \left(\frac{\sum\alpha}{3}, \frac{\sum\alpha}{3}, \frac{\sum\alpha}{3}\right) = \left(-\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\right)$ which lies on x = y = z.

9. If a > b > c and $a^2 = b = \sqrt{c}$ then $b < 1 \Rightarrow \cot^{-1}x < 1 \Rightarrow x > \cot 1$

10. $a \le 4\sin A \Rightarrow \frac{a}{\sin A} \le 4 \Rightarrow R \le 2$
so for any point (x, y) inside the circumcircle, $x^2 + y^2 < 4$
$\Rightarrow |xy| < 2$

11. a, b and $\angle A$ are given in ΔABC,
$\therefore \cos A = \frac{b^2 + c^2 - a^2}{2bc}$
$\Rightarrow c^2 - 2bc\cos A + (b^2 - a^2) = 0$
$\Rightarrow c^2 - (2b\cos A)c + (b^2 - a^2) = 0$
which is quadratic in c and gives two values of c and let these are c_1 and c_2
$\therefore c_1 + c_2 = 2b\cos A$
and $c_1 \cdot c_2 = (b^2 - a^2)$
$c_1^2 + c_2^2 - 2c_1c_2\cos 2A$
$= (c_1 + c_2)^2 - 2c_1c_2(1 + \cos 2A)$
$= 4b^2\cos^2 A - 2(b^2 - a^2)(2\cos^2 A)$
$= 4b^2\cos^2 A - 4b^2\cos^2 A + 4a^2\cos^2 A$
$= 4a^2\cos^2 A$

12. $\frac{1}{\sqrt{2}+\sqrt{5}} + \frac{1}{\sqrt{5}+\sqrt{8}} + \frac{1}{\sqrt{8}+\sqrt{11}} + \cdots$ n terms
$= \frac{\sqrt{5}-\sqrt{2}}{3} + \frac{\sqrt{8}-\sqrt{5}}{3} + \frac{\sqrt{11}-\sqrt{8}}{3} + \cdots + \frac{\sqrt{5+(n-1)3} - \sqrt{2+(n-1)3}}{3}$
$= \frac{\sqrt{3n+2}-\sqrt{2}}{3} = \frac{3n+2-2}{3\left(\sqrt{3n+2}+\sqrt{2}\right)} = \frac{n}{\sqrt{3n+2}+\sqrt{2}}$

$= \frac{n}{\sqrt{2+3n}+\sqrt{2}} < \frac{n}{\sqrt{3n}} < n$

13. $\sin\frac{\pi}{24} = \frac{2}{R-2}$

$\Rightarrow R = 2\left(1+\operatorname{cosec}\frac{\pi}{24}\right)$

(A) is true $\because$ $\frac{\left(\sin\frac{\alpha}{2}+\cos\frac{\alpha}{2}\right)^2}{\sin A} = 1+\operatorname{cosec}\alpha$

(D) is true

14. $x^2 + x + 1 = |[x]|$
Hence only 2 solutions exist
i.e. $x = -1$ and another lie $-2 < x < -1$

15. $\sin x + \cos x = 1 - a \sin x \cos x$
$\Rightarrow a^2 \sin^2 x \cos^2 x - 2(a + 1) \sin x \cos x = 0$
$\Rightarrow \sin 2x\left(\frac{a^2}{2}\sin 2x - 2(a+1)\right) = 0$
Hence $\sin 2x = 0 \ \forall\ a \in R$
$\Rightarrow x = \frac{n\pi}{2}$, $n \in I$
And $\sin 2x = \frac{4(a+1)}{a^2} \Rightarrow a \in (-\infty, 2 - 2\sqrt{2}\,] \cup [2 + 2\sqrt{2}\,, \infty)$

SECTION – C

1. $a = \sum_{r=1}^{\infty}\frac{1}{r^2},\ b = \sum_{r=1}^{\infty}\frac{1}{(2r-1)^2}$

$a = \frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\ldots\infty$ and $b = \frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\frac{1}{7^2}+\ldots\infty$

$b = \left(\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\ldots\infty\right)-\left(\frac{1}{2^2}+\frac{1}{4^2}+\frac{1}{6^2}+\ldots\infty\right)$

$b = \left(\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\ldots\infty\right)-\frac{1}{2^2}\left(1+\frac{1}{2^2}+\frac{1}{3^2}+\ldots\infty\right)$

$b = a - \frac{1}{4}a$

$b = a \times \frac{3}{4}$

$\frac{a}{b} = \frac{4}{3}$

2. $\int_0^{\pi} f^{-1}(x)dx = \int_0^{\pi} t f'(t)dt$

$= [t\, f(t)]_0^{\pi} - \int_0^{\pi} (t + \sin t)\, dt$

$= \pi^2 - \frac{\pi^2}{2} - 1 - 1$

$= \frac{\pi^2}{2} - 2$

$\Rightarrow k = 2$

3. **Case I:** When we use 6, 7 or 8 at ten thousand place then number of numbers
$= 3 \times {}^4P_4 = 72$
Case II: When we use 5 at ten thousand place and 6, 7 or 8 at thousand place then number of numbers is
$= 1 \times 3 \times {}^3P_3 = 18$
hence the required numbers of numbers is 72 + 18 = 90

4. f(x) is an odd function
$\Rightarrow$ g(x) is $f^{-1}(x)$
$\Rightarrow$ Area $= \int_0^{2\pi} f^{-1}(y)\, dy = \int_0^{2\pi} x\, f'(x)\, dx$
$= \left[x(f(x))\big|_0^{2\pi} - \int_0^{2\pi} (x + \sin x)\, dx \right] = 2\pi^2$

5. $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 2(a^2 + b^2 + c^2) - 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$
$= 2(4 + 9 + 16) - 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$
$= 58 - 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$.
Now $(\vec{a} + \vec{b} + \vec{c})^2 \geq 0$
$\Rightarrow a^2 + b^2 + c^2 \geq -2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$
$\Rightarrow -2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \leq 29$
$\Rightarrow |\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 \leq 87$